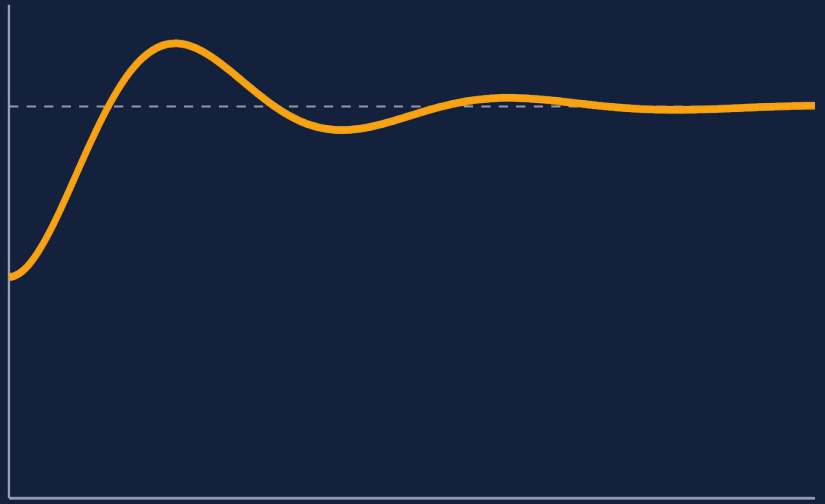
Steady-State Error and Error Constants



Control Systems (22EEG53)

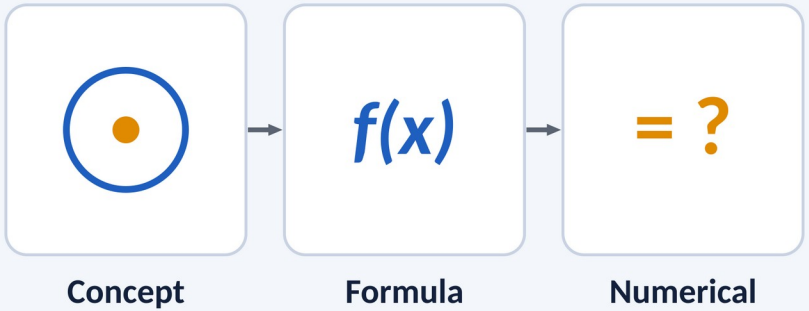
Unit 2: Time Domain Analysis of Control Systems

Topic: Type of systems, static error constants, steady-state error, generalized error series

Prof. Mitesh Astik

Learning Outcomes

- Find the type of a system from G(s).
- Calculate Kp, Kv, Ka and the steady-state error.
- Predict error for step, ramp and parabolic inputs.
- Use the generalized error series for general inputs.



Linked to CO2
Analyse response during transient and steady-state conditions.

Recap: What You Already Know

- Unity feedback: error e = r − c.
- Closed-loop stable systems have a final value.
- Final value theorem: lim s·F(s) as s → 0.
- Test signals: 1/s, $1/s^2$, $1/s^3$.



E(s) = R(s) / (1 + G(s))

Final value theorem:

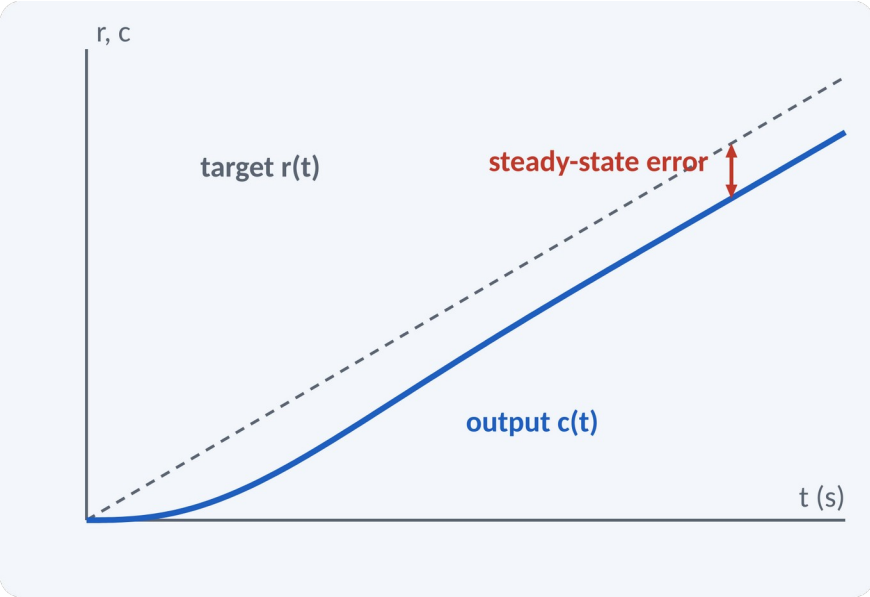
ess = lim s→0 s·R(s) / (1 + G(s))

valid only if the closed loop is stable

lim t→∞ f(t) = lim s→0 s·F(s) (poles in left half-plane only)

Motivation: Tracking Accuracy

- A radar must follow a moving aircraft.
- A robot must follow a path precisely.
- Transient dies out but a constant gap may remain.
- Steady-state error measures that final gap.



Design aim: keep ess within the allowed limit for the expected input

Steady-State Error

- Error: e(t) = r(t) − c(t) (unity feedback).
- E(s) = R(s)/(1 + G(s)).
- Apply the final value theorem to E(s).
- The result depends on G(s) at s → 0.
- Valid only for a stable closed loop.



E(s) = R(s) / (1 + G(s))

Final value theorem:

ess = lim s→0 s·R(s) / (1 + G(s))

valid only if the closed loop is stable

ess = lim s→0 s·R(s)/(1 + G(s))
G(s) open-loop transfer function, R(s) input

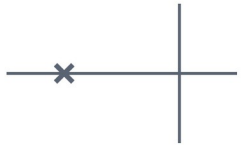
Type of a System

- Type = number of poles of G(s) at s = 0.
- Type 0: no integrator.
- Type 1: one integrator.
- Type 2: two integrators.
- Type tells which inputs are tracked with zero error.

Type 0

$K / [(\tau 1\, s + 1)(\tau 2\, s + 1)]$

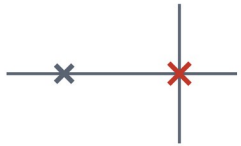
no pole at origin



Type 1

$K / [s\,(\tau\, s + 1)]$

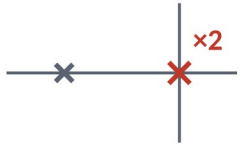
one pole at origin



Type 2

$K / [s^2\,(\tau\, s + 1)]$

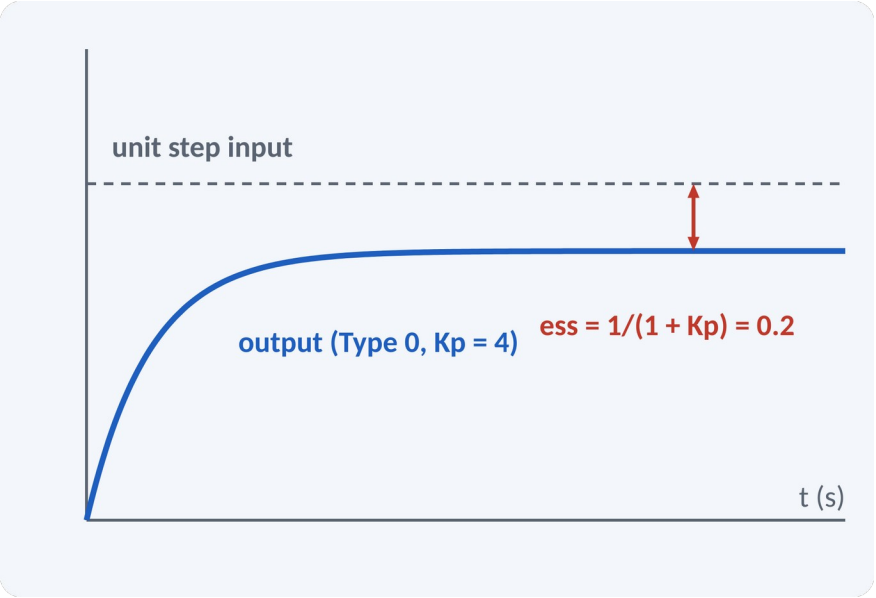
two poles at origin



$$G(s) = K \prod(\tau_i\, s + 1) / [s^N \prod(\tau_j\, s + 1)];\quad \text{type} = N$$

Static Position Error Constant Kp

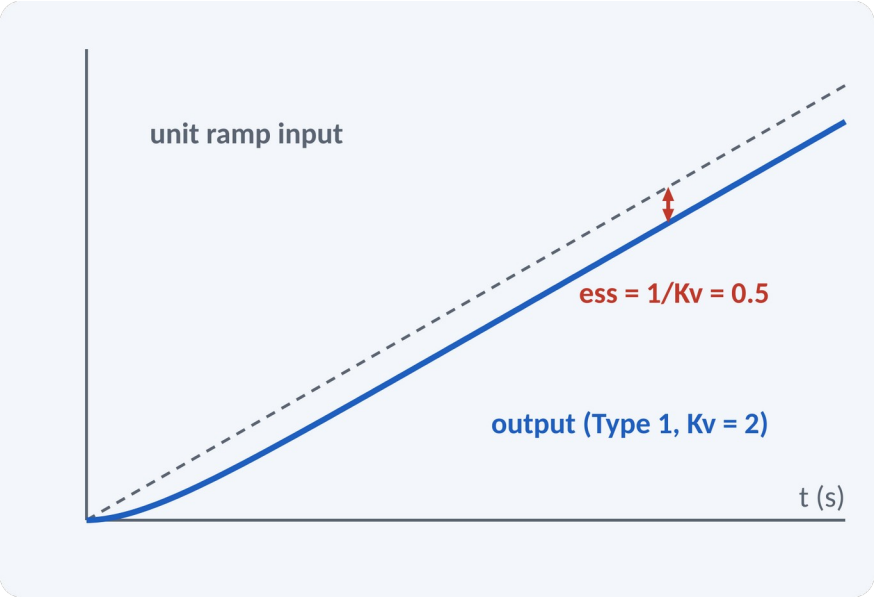
- Input: unit step, R(s) = 1/s.
- Kp = lim G(s) as s → 0.
- Type 0: Kp = K, finite error.
- Type 1 or higher: Kp = ∞, zero error.



ess = 1/(1 + Kp) for a unit step; A/(1 + Kp) for amplitude A
Kp = lim s→0 G(s)

Static Velocity Error Constant Kv

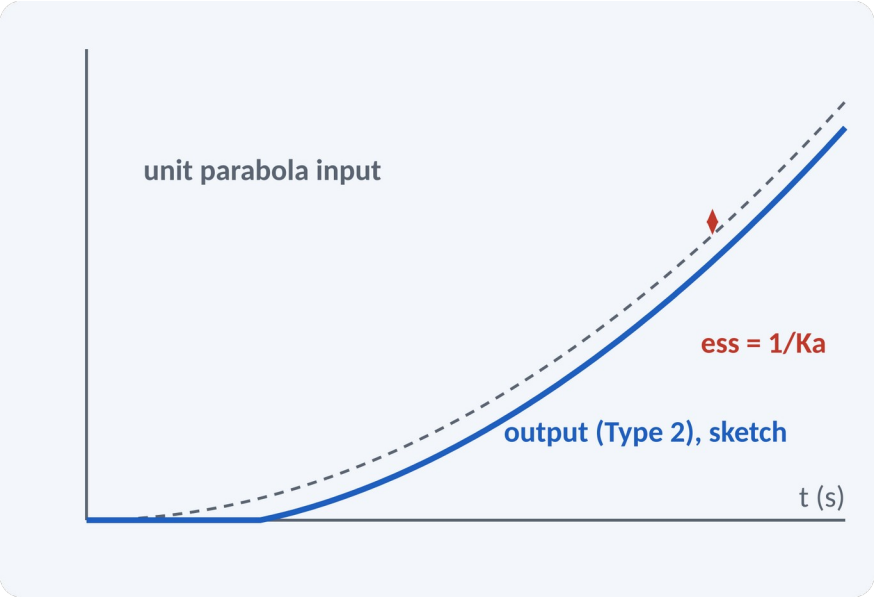
- Input: unit ramp, $R(s) = 1/s^2$.
- $Kv = \lim s \cdot G(s)$ as $s \to 0$.
- Type 0: ess = ∞.
- Type 1: finite error 1/Kv.
- Type 2 or higher: zero error.



ess = A/Kv for a ramp of slope A
$Kv = \lim_{s \to 0} s \cdot G(s)$

Static Acceleration Error Constant Ka

- Input: parabola $A \cdot t^2/2$, $R(s) = A/s^3$.
- $Ka = \lim s^2 \cdot G(s)$ as $s \to 0$.
- Types 0 and 1: infinite error.
- Type 2: finite error A/Ka.
- Type 3 or higher: zero error.



$ess = A/Ka; \quad Ka = \lim s \to 0\ s^2 \cdot G(s)$

Error Table: Type vs Input

- Higher type → better tracking, but harder stability.
- Zero error needs enough integrators.
- Finite errors use Kp, Kv or Ka.
- Superposition applies for combined inputs.

	Step	Ramp	Parabola
Type 0	A/(1+Kp)	∞	∞
Type 1	0	A/Kv	∞
Type 2	0	0	A/Ka

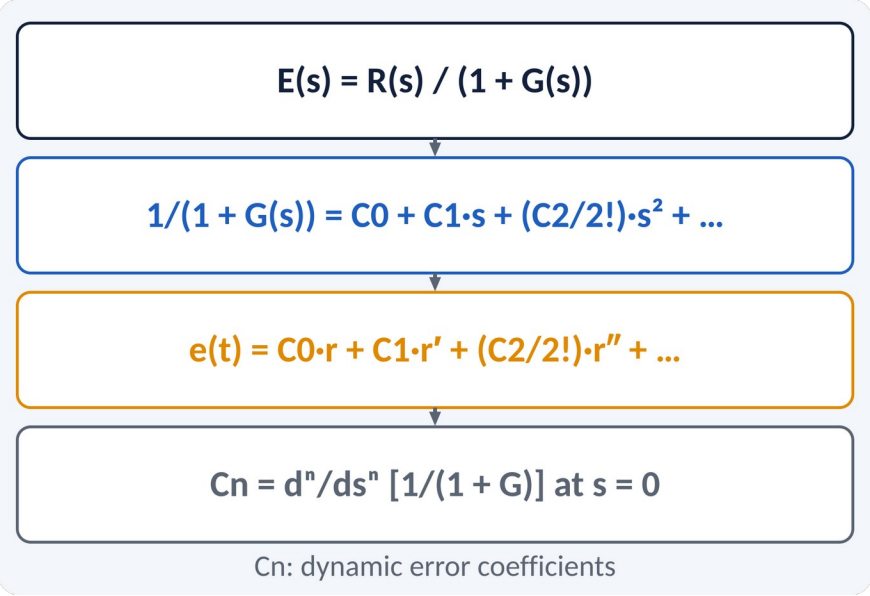
$$Kp = \lim G(s) \quad Kv = \lim s \cdot G(s) \quad Ka = \lim s^2 \cdot G(s)$$

(limits as $s \to 0$; A is the input amplitude)

$$r = A \cdot u(t) + B \cdot t + (C/2) \cdot t^2 \rightarrow ess = A/(1+Kp) + B/Kv + C/Ka$$

Generalized Error Series

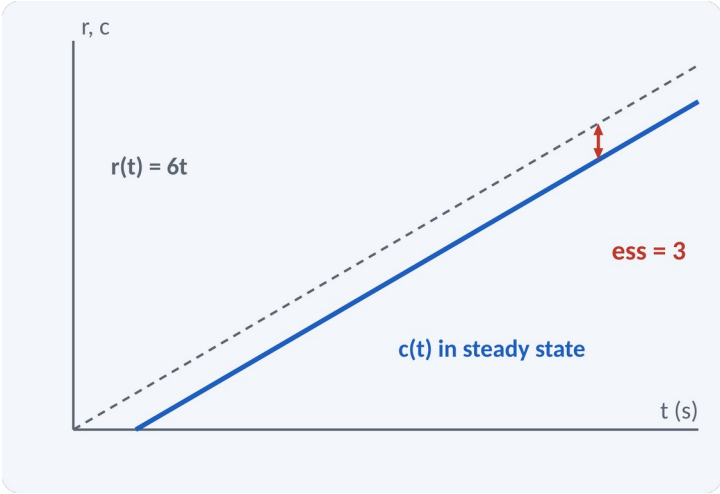
- Static constants cover only step, ramp, parabola.
- Series handles any smooth input r(t).
- Expand 1/(1 + G(s)) in powers of s.
- Coefficients Cn are dynamic error coefficients.
- Error follows from the derivatives of r(t).



e(t) = C0·r(t) + C1·r′(t) + (C2/2!)·r″(t) + ...
Cn = dⁿ/dsⁿ [1/(1 + G(s))] at s = 0

Solved Example 1: Static Constants

- Unity feedback, G(s) = 20(s + 2)/[s(s + 4)(s + 5)]. Input r = 6t. Find type, Kp, Kv, Ka, ess.
- 1. Stability: $s^3 + 9s^2 + 40s + 40$ → all Routh entries positive: stable.
- 2. One pole at origin: Type 1.
- 3. Kp = ∞.
- 4. Kv = lim sG = 20×2/(4×5) = 2.
- 5. Ka = lim s^2G = 0.
- 6. Ramp slope A = 6: ess = A/Kv = 3.



Answer: Type 1, Kp = ∞, Kv = 2, Ka = 0, ess = 3 for r = 6t

Solved Example 2: Error Series

- Unity feedback, $G(s) = 10(s + 1)/s^2$. Input $r(t) = 3t^2$. Find ess by Ka and by the error series.
- 1. Type 2: $Ka = \lim s^2G = 10(0 + 1) = 10$.
- 2. $r = 3t^2 = 6 \cdot t^2/2$, so A = 6: $ess = 6/10 = 0.6$.
- 3. Series: $1/(1+G) = s^2/(s^2 + 10s + 10) = 0.1s^2 - 0.1s^3 + \ldots$
- 4. C0 = C1 = 0, C2/2! = 0.1.
- 5. $e = 0.1 \cdot r'' = 0.1 \times 6 = 0.6$.

$G(s) = 10(s + 1)/s^2$ (Type 2)

$1/(1 + G) = s^2/(s^2 + 10s + 10)$

$= 0.1s^2 - 0.1s^3 + \ldots$

$C0 = 0, C1 = 0, C2 = 0.2$

$r = 3t^2 \rightarrow r'' = 6 \rightarrow ess = (C2/2!) \cdot 6 = 0.6$

Answer: ess = 0.6 by both methods

Common Mistakes and Exam Tips

Common mistakes

- Applying the theorem to an unstable closed loop.
- Counting type from the closed-loop transfer function.
- Forgetting to write G(s) in time-constant form for K.
- Using the amplitude of r(t) wrongly (A for $t^2/2$).
- Using static constants for non-unity feedback directly.

Exam tips

- Check stability first (Routh).
- Count poles of G(s) at s = 0 only.
- Write K as the gain in time-constant form.
- Convert to unity feedback if H(s) ≠ 1.

Summary

- ess = lim s·R(s)/(1 + G(s)) for stable unity feedback.
- Type = number of integrators in G(s).
- Kp = lim G, Kv = lim sG, Ka = lim s^2G.
- Errors: A/(1+Kp), A/Kv, A/Ka for step, ramp, parabola.
- Error series: e(t) = ΣCn $r^{(n)}$/n!.

	Step	Ramp	Parabola
Type 0	A/(1+Kp)	∞	∞
Type 1	0	A/Kv	∞
Type 2	0	0	A/Ka

Kp = lim G(s) Kv = lim s·G(s) Ka = lim s^2·G(s)

(limits as s → 0; A is the input amplitude)

Zero / finite / infinite error follows the type–input table

Quick Quiz

Q1. G(s) = 10/[s(s+2)] is of type:
(a) 0 (b) 1 (c) 2 (d) 3

Ans: (b)

Q2. Kv of G(s) = 10/[s(s+2)] is:
(a) 0 (b) 2 (c) 5 (d) ∞

Ans: (c)

Q3. Ramp error of a Type 0 system is:
(a) 0 (b) finite (c) 1/Kp (d) infinite

Ans: (d)

Q4. Unit step error of Type 0 with Kp = 4 is:
(a) 0.2 (b) 0.25 (c) 4 (d) 0

Ans: (a)

Q5. Ka = lim of:
(a) G(s) (b) sG(s) (c) $s^2G(s)$ (d) G(s)/s

Ans: (c)

References

1. **I. J. Nagrath and M. Gopal, Control Systems Engineering**
   5th Ed., New Age International (P) Ltd., 2008 (Units 1, 2 textbook)

2. **K. Ogata, Modern Control Engineering**
   5th Ed., Pearson Education Asia / PHI, 2009 (reference)

3. **P. S. Satyanarayana, Concepts of Control Systems**
   2nd Ed., Dynaram Publishers, Bangalore, 2001 (reference)

Next lecture: Unit 3: frequency-domain specifications and Bode plots